University of Stuttgart
Institute of electrical and optical communications

Advanced optical fiber communications Wintersemester 2025/6 Exercise 3

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Recap:

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- Polarization of plane electromagnetic waves
- Dispersion: Group velocity is the energy velocity for electromagnetic waves
- Dispersion: Sellmeier equation, refractive index, group index, dispersion coefficient

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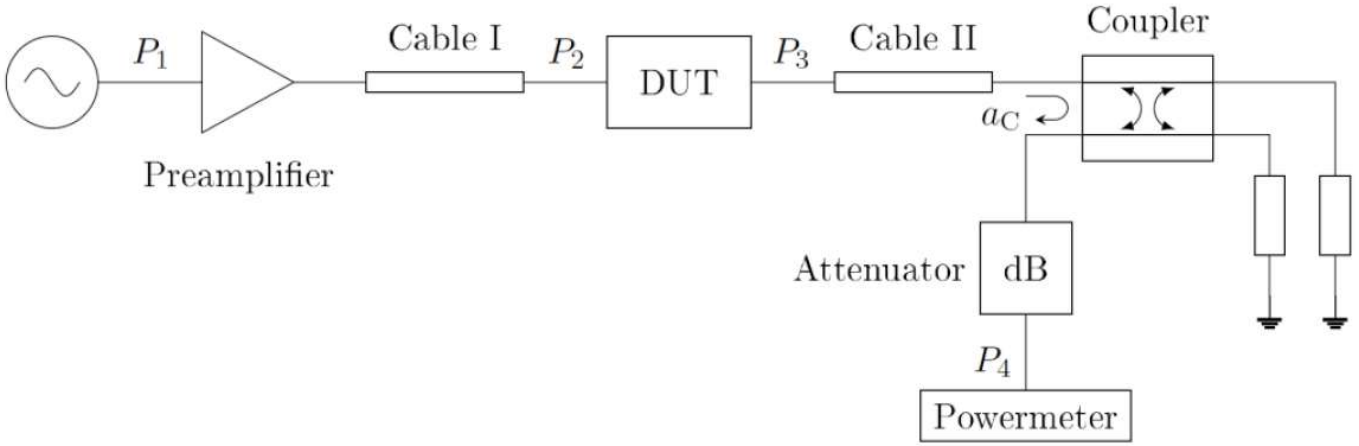


Figure 1 Schematic of the measurement setup

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For measuring RF power amplifiers, a measurement setup based on fig. 1 is used. The following table gives the characteristics of the setup.

Max. power of the signal generator	$P_{sig,max} = 20$ mW
Gain of the preamplifier	$G_{pre} = 400$
Max. input power of the preamplifier	$P_{1,max} = 16$ mW
Coupling loss	$a_C = 20$ dB
Loss of the attenuator	$a_A = 22$ dB
Max. input power of the powermeter	$P_{4,max} = 45$ mW
Gain of the device under test (DUT)	$G_{DUT} = 10\ldots40$

a) Convert all powers from the table into dBm and all gains into dB.

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In this setup, the cables and connectors cause unwanted attenuation, which need to be found with a calibration routine in order to make precise measurements of the DUT. The routine consists of two measurements, while the DUT is removed from the setup: At first, the power P_2 is measured with a known power P_1. Then, the nodes of P_2 and P_3 are directly connected and P_4 is evaluated.

b) Measurement 1: The power of the signal generator is $P_{sig} = P_1 = 5$ µW. The measured power is $P_2 = 1.8$ mW. What is the cable loss of cable I a_I?

c) Measurement 2: The measurement results in $P_4 = 100$ nW. Find the loss a_{II} of cable II.

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Now, a power amplifier as DUT is being measured.

d) The input power at the DUT has to be $P_2 = 1$ W. What is the power at the signal generator (in dBm and in mW) that has to be fed to the preamplifier considering the calibration?

e) The power meter measures $P_4 = 0.8$ dBm. What is the output power P_3 of the DUT? Find the gain G_{DUT} in dB. How much is the power P_2 amplified by the DUT?

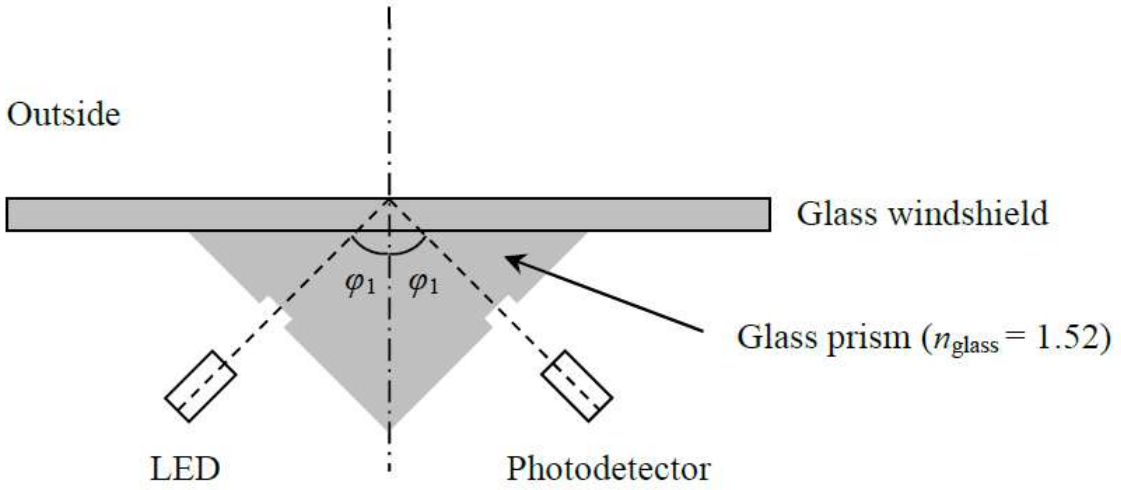
Problem 5:

An electromagnetic wave propagates in a waveguide, e.g. an optical fiber, with a loss of 0.4 dB/km. Find the distances that the wave has to travel until the power decays to 1/2 (1/10, 1/100) of its initial value.

Problem 6:

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A simplified graph of a rain sensor is illustrated in the figure below. The rain sensor is fixed on the glass windshield of a car ($n_{glass} = 1.52$). The LED and the photodetector are positioned so that $\varphi_1 = 45°$.



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a) Calculate the critical angle for the case of a dry windshield and sketch the ray trace of the light from the LED notch to the photodetector notch. What is the power reflection coefficient for the perpendicular and parallel polarization?

b) Sketch the ray trace in the case of a wet windshield on a rainy day ($n_w = 1.33$). Calculate the power reflection coefficients of both polarizations.

c) What is the further procedure to improve the sensitivity of the rain sensor?

d) What would be the incident angle of the LED light, if no parallel polarization has to be detected by the photodetector on a rainy day?

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Thank you!

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